

Lecture 11b

EE-215 Electronic Devices and Circuits

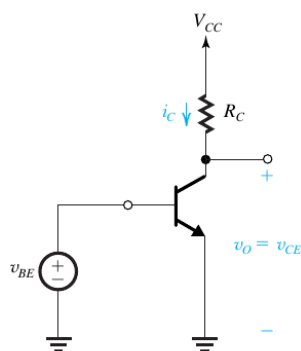
Asst Prof Muhammad Anis Ch

Applying the BJT in Amplifier Design

- When operated in active mode, the BJT functions as a voltage-controlled current source.
 - (i.e. the base-emitter voltage v_{BE} controls the collector current i_C)
 - this enables us to use the BJT in the design of amplifiers

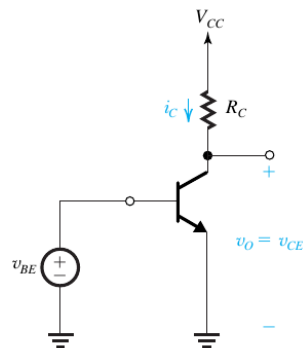
Obtaining a Voltage amplifier

- A voltage controlled current source can be used as a transconductance amplifier
 - A transconductance amplifier is an amplifier whose input signal is a voltage and output signal is a current
 - we usually are more interested in a voltage amplifier
 - A transconductance amplifier can be converted into a voltage amplifier,
 - by passing the output current through a resistor
 - and taking the voltage across the resistor as output

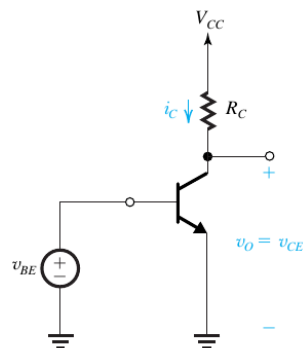


- in the figure
 - v_{BE} is the input voltage
 - R_C is called the load resistance.
 - R_C converts the collector current i_C to a voltage $i_C R_C$
 - V_{CC} is the supply voltage
 - here the output voltage is taken between the collector terminal and the ground (rather than across R_C)

- this arrangement enables us to maintain a ground reference through out the circuit
- $\Rightarrow$
- v_{BE} is the input voltage
- v_{CE} is the output voltage



- Using KVL, an expression for v_{CE} can be computed
 - i.e. $V_{CC} = i_C R_C + v_{CE}$
 - or $v_{CE} = V_{CC} - i_C R_C$
 - $\Rightarrow$ output voltage v_{CE} is an inverted version of $i_C R_C$ that is shifted by the constant value of the supply voltage V_{CC}



The Voltage Transfer Characteristic (VTC)

- VTC is a plot of the output voltage versus the input voltage
 - and it can provide a better insight in to the operation of an amplifier circuit

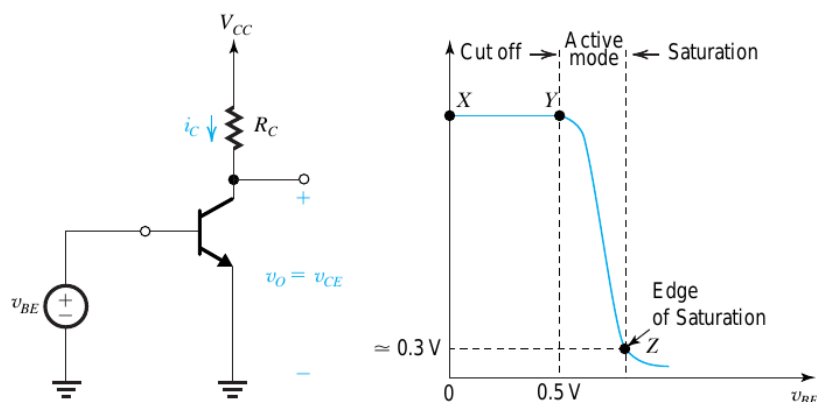
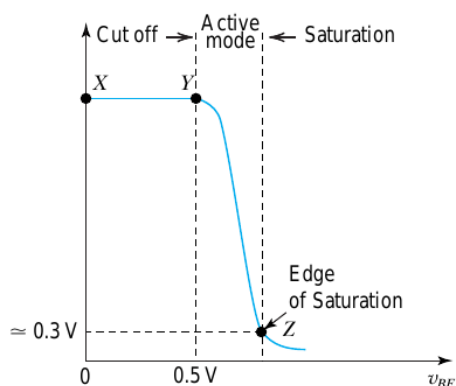
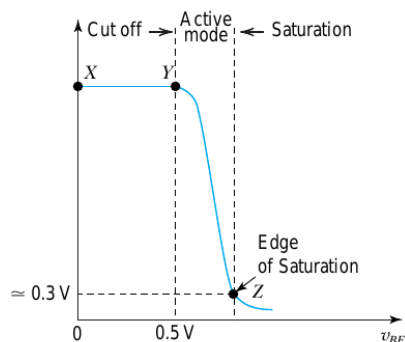


Figure 6.31 (a) Simple BJT amplifier with input v_{BE} and output v_{CE} . (b) The voltage transfer characteristic (VTC) of the amplifier in (a). The three segments of the VTC correspond to the three modes of operation of the BJT.

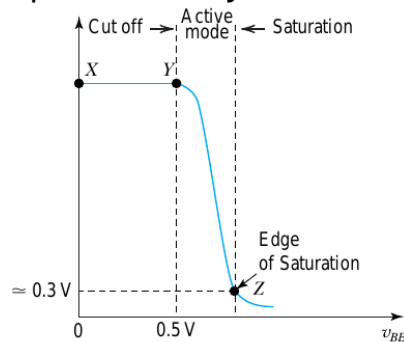
- Note that for $v_{BE} < 0.5V$, the transistor is cut-off $\Rightarrow i_C = 0$
 - and $v_{CE} = V_{CC} - i_C R_C = V_{CC} - (0)R_C = V_{CC}$
 - As v_{BE} exceeds $0.5V$, the transistor turns on and v_{CE} decreases.
 - As initially i_C is still small
 - $\Rightarrow v_{CE}$ is still high
 - $\Rightarrow$ BJT is operating in active region (active region: $v_{CE} \geq 0.3V$)



- BJT stays in active region as v_{BE} is increased
 - until the value of v_{BE} is reached for which v_{CE} becomes lower than v_{BE} by $0.4V$ i.e. $v_{CE} \approx 0.3V$
 - this is point Z



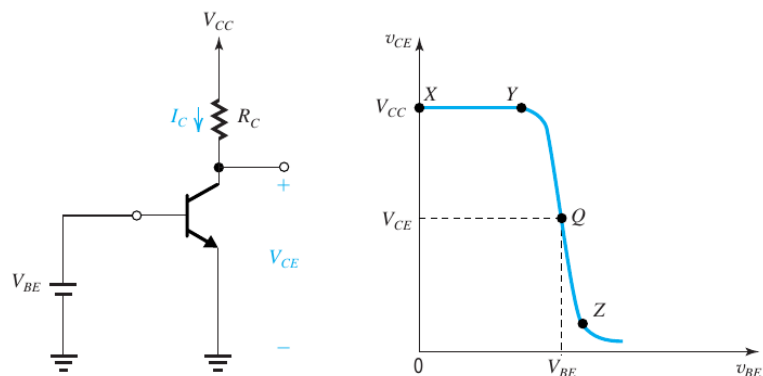
- for $v_{CE} < 0.3V$ (i.e. beyond point Z), the transistor enters the saturation region,
- from the fig, it is clear that the segment of greatest slope
 - (and hence the largest amplifier gain) is that labeled as YZ
 - which corresponds to the active mode of operation
 - thus for amplifier application, one will always choose to operate the BJT in active mode



- thus for amplifier application, the BJT will be operated in active mode i.e. in the segment YZ
 - for active mode operation (ignoring base-width modulation or the Early Effect)
 - $i_C = I_S e^{v_{BE}/V_T}$
 - As $v_{CE} = V_{CC} - i_C R_C$
 - $\Rightarrow v_{CE} = V_{CC} - (I_S e^{v_{BE}/V_T}) R_C$
 - $v_{CE} = V_{CC} - R_C I_S e^{v_{BE}/V_T}$
 - Although this is a non-linear relationship, but linear (or almost linear) amplification is possible,
 - by using the technique of biasing the BJT

Biasing the BJT to Obtain Linear Amplification

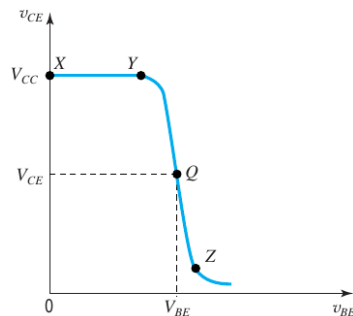
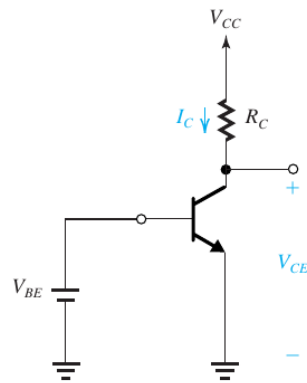
- Biasing enables us to obtain almost-linear amplification from the BJT



- **Figure 6.32** Biasing the BJT amplifier at a point Q located on the active-mode segment of the VTC.
 - A dc voltage V_{BE} is selected to obtain operation at a point Q

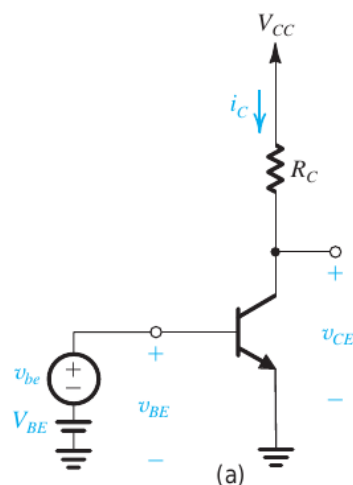
on the segment YZ

- the resulting V_{CE} can be given as
 - $V_{CE} = V_{CC} - R_C I_S e^{V_{BE}/V_T}$
 - Point Q is called the bias point
 - or the dc operating point
 - or the Quiescent point

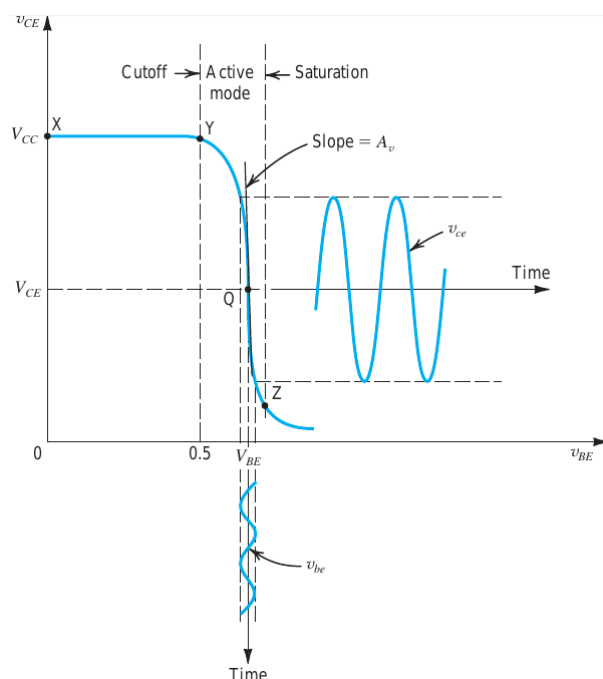


- Once the Q point is selected, the ac signal to be amplified is superimposed on the bias voltage V_{BE}
 - $\Rightarrow$ the total instantaneous value of v_{BE} is
 - $v_{BE}(t) = V_{BE} + v_{be}(t)$
 - the resulting v_{CE} is

- $v_{CE} = V_{CC} - R_C I_S e^{v_{BE}/V_T}$

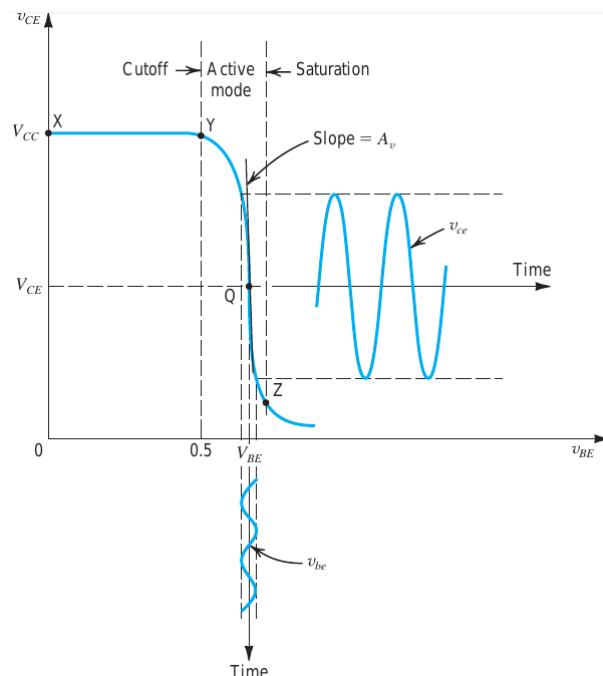


- $v_{CE}(t)$ can also be obtained graphically from the VTC
 - Note that if the amplitude of v_{be} is small enough to
 - restrict the signal swing to a short almost-linear segment
 - of the VTC, linear amplification is achieved
 - The shorter the segment, the greater the linearity achieved.



The Small-Signal Voltage Gain

- if the input signal v_{be} is kept small, the output signal v_{ce} is nearly proportional to v_{be}
 - i.e. $v_{ce} \propto v_{be}$
 - or $v_{ce} = A_v v_{be}$
 - A_v is the proportionality constant and is the voltage gain of the amplifier
 - A_v is the slope of the almost linear segment of the VTC around Q
 - i.e. $A_v = \left. \frac{\partial v_{CE}}{\partial v_{BE}} \right|_{v_{BE}=V_{BE}}$



$$\bullet A_v = \left. \frac{\partial v_{CE}}{\partial v_{BE}} \right|_{v_{BE}=V_{BE}}$$

$$\circ \text{ using } v_{CE} = V_{CC} - R_C I_S e^{v_{BE}/V_T}$$

$$\blacksquare A_v = \left. \frac{\partial}{\partial v_{BE}} [V_{CC} - R_C I_S e^{v_{BE}/V_T}] \right|_{v_{BE}=V_{BE}}$$

$$\blacksquare A_v = \left. \left[0 - R_C I_S \frac{\partial}{\partial v_{BE}} (e^{v_{BE}/V_T}) \right] \right|_{v_{BE}=V_{BE}}$$

$$\blacksquare A_v = \left. \left[-R_C I_S e^{v_{BE}/V_T} \frac{\partial}{\partial v_{BE}} (v_{BE}/V_T) \right] \right|_{v_{BE}=V_{BE}}$$

$$\blacksquare A_v = \left. \left[-\frac{R_C I_S}{V_T} e^{v_{BE}/V_T} \right] \right|_{v_{BE}=V_{BE}}$$

$$\blacksquare A_v = -\frac{R_C I_S}{V_T} e^{V_{BE}/V_T} = -\frac{R_C}{V_T} I_S e^{V_{BE}/V_T}$$

$$\blacksquare A_v = -\frac{R_C}{V_T} I_C \because I_C = I_S e^{V_{BE}/V_T}$$

$$\blacksquare \Rightarrow A_v = -\frac{I_C}{V_T} R_C$$

$$\bullet A_v = -\frac{I_C}{V_T} R_C$$

◦ from this expression, Note that

1. The voltage gain is negative

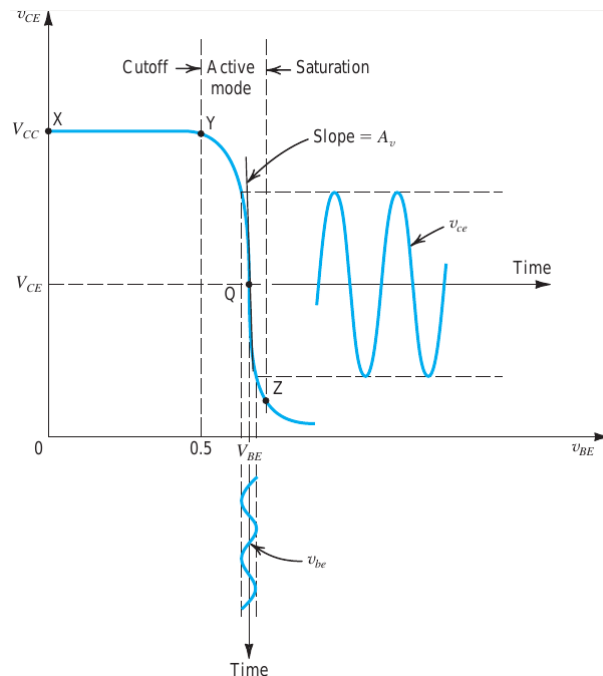
■ $\Rightarrow$ the amplifier is inverting.

■ i.e. there is a 180° phase shift between the input and the output

■ this inversion is clear from the figure

2. The gain is proportional to the load resistance R_C ,

■ and to the collector bias current I_C



- $A_v = -\frac{I_C}{V_T} R_C$
 - $A_v = -\frac{I_C R_C}{V_T} = -\frac{V_{RC}}{V_T}$
 - where V_{RC} is the dc voltage drop across R_C i.e.
 $V_{RC} = V_{CC} - V_{CE}$
 - thus the gain is simply the ratio of the dc voltage drop across the load resistor R_C
 - to the thermal voltage V_T ($\approx 25mV$ at room temperature)
 - this relation is useful in determining the absolute upper limit on the magnitude of voltage gain for this amplifier circuit
- Note that
 - the theoretical maximum gain A_v is obtained by biasing the BJT at the edge of saturation,
 - (which would not leave any room for negative signal swing)
 - The resulting gain can be given as
 - $A_v = -\frac{V_{RC}}{V_T} = -\frac{V_{CC} - V_{CE}}{V_T} = -\frac{V_{CC} - 0.3}{V_T} \because V_{CE} = 0.3$ at edge of saturation
 - As $V_{CC} - 0.3 \approx V_{CC}$
 - $A_{vmax} \approx -\frac{V_{CC}}{V_T}$ or $|A_{vmax}| \approx \frac{V_{CC}}{V_T}$

Determining the VTC by Graphical Analysis

- Voltage transfer characteristic (VTC) of the amplifier can be determined using graphical method

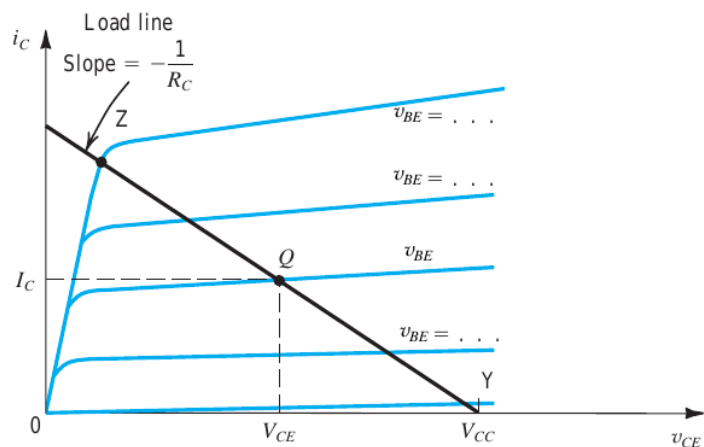


Figure 6.34 Graphical construction for determining the VTC of the amplifier circuit of Fig. 6.33(a).

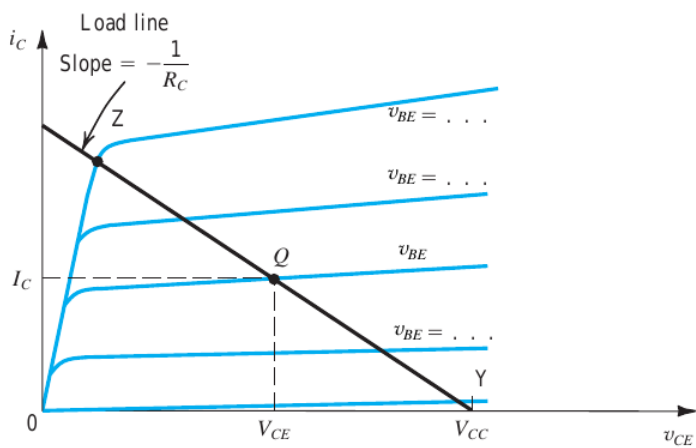
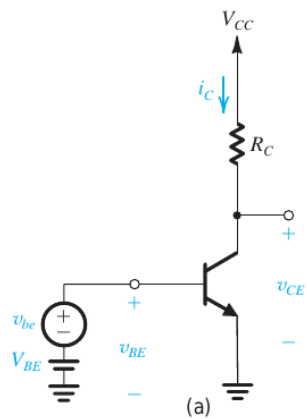
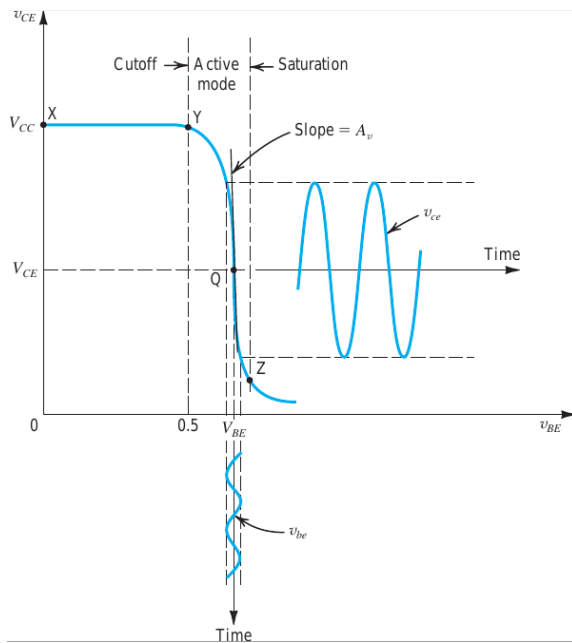


Figure 6.34 Graphical construction for determining the VTC of the amplifier circuit of Fig. 6.33(a).



- the graphical analysis is based on the fact that
 - for each value of v_{BE} , the circuit will be operating
 - at the point of intersection of the $i_C - v_{BE}$ graph
 - corresponding to that particular value of v_{BE}
 - and the straight line representing the KVL eq
 - i.e. $v_{CE} = V_{CC} - R_C i_C \Rightarrow i_C = \frac{V_{CC}}{R_C} - \frac{v_{CE}}{R_C}$

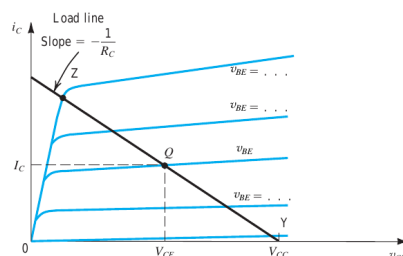


Figure 6.34 Graphical construction for determining the VTC of the amplifier circuit of Fig. 6.33(a).

- The straight line representing this relationship is superimposed on the $i_C - v_{CE}$ characteristic
- this line intersects the horizontal axis at $v_{CE} = V_{CC}$ and has a slope of $-\frac{1}{R_C}$
 - this line is called the load line as it represents the load resistance R_C
 - the VTC can then be determined point by point
 - here three important points have been labeled
 - At point Y
 - $v_{BE} = 0.5V$ (cutin voltage of EBJ)

■ At point Q

- BJT can be biased for amplifier application i.e.

$$v_{BE} = V_{BE} , v_{CE} = V_{CE}$$

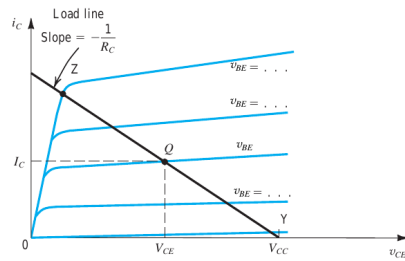


Figure 6.34 Graphical construction for determining the VTC of the amplifier circuit of Fig. 6.33(a).

• At point Z,

- the BJT leaves the active region and enters the saturation region

- Note that if the BJT is to be used as a switch, then operating points Y and Z are applicable

- At Y, the transistor is off (open switch)
- and at Z the transistor operates as a low valued resistance R_{CEsat} and has a small voltage drop (closed switch)

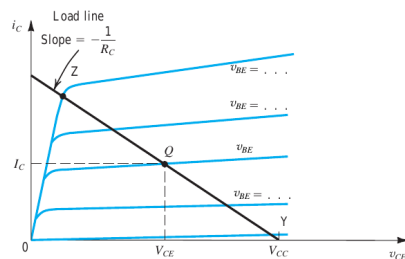


Figure 6.34 Graphical construction for determining the VTC of the amplifier circuit of Fig. 6.33(a).

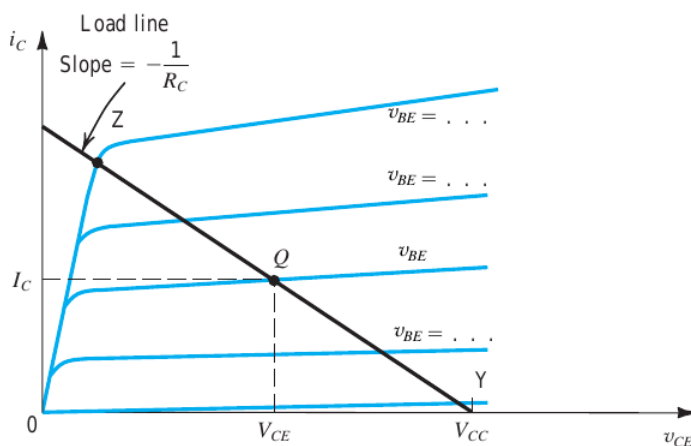
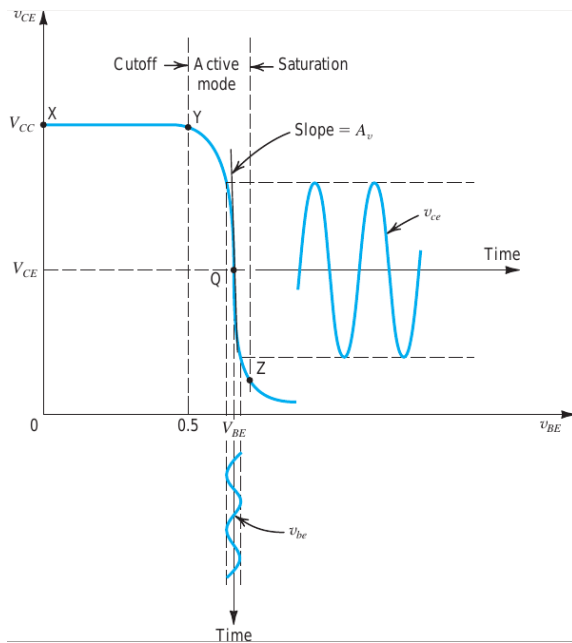


Figure 6.34 Graphical construction for determining the VTC of the amplifier circuit of Fig. 6.33(a).



Locating the Bias Point Q

- The bias point Q is determined by the value of v_{BE} and that of the load resistance R_C
 - Two important considerations in choosing the location of Q are
 - the required gain and
 - the allowable signal swing at the output

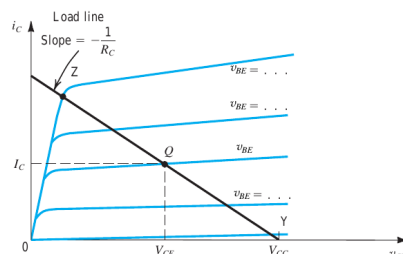
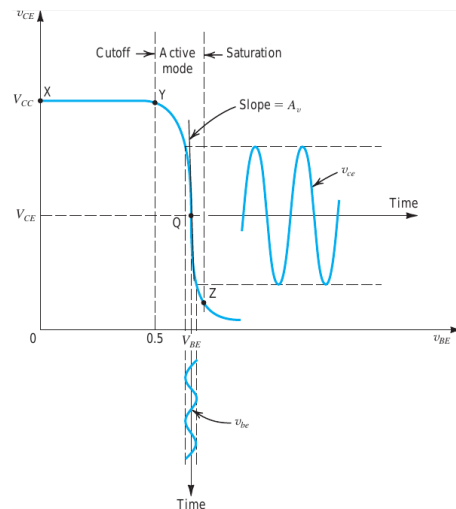
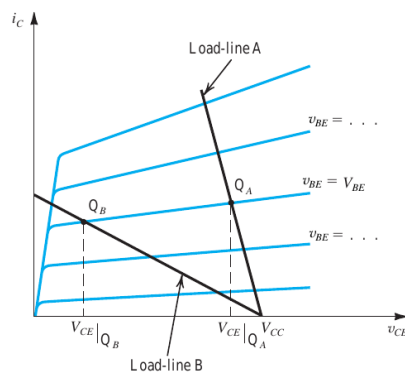


Figure 6.34 Graphical construction for determining the VTC of the amplifier circuit of Fig. 6.33(a).

- Consider the VTC shown in figure
 - here the value of R_C is fixed
 - the only remaining variable is the value of V_{BE}
 - Note that the slope increases as we move closer to point Z
 - thus higher gain is achieved by locating Q as close to Z as possible
 - However the closer Q is to the boundary point Z,
 - the smaller the allowable magnitude of negative signal swing



- to decide on the value of R_C , $i_C - v_{CE}$ plane is useful
 - here two load lines are shown which result in two extreme bias points
 - Point Q_A is too close to V_{CC}
 - resulting in a severe constraint on the positive signal swing of v_{ce}
 - Exceeding the allowable +ve maximum will result in positive peaks of the signal being clipped off,
 - since the BJT will turn off for the part of each cycle near the +ve peak
 - we say the circuit does not have sufficient "headroom"



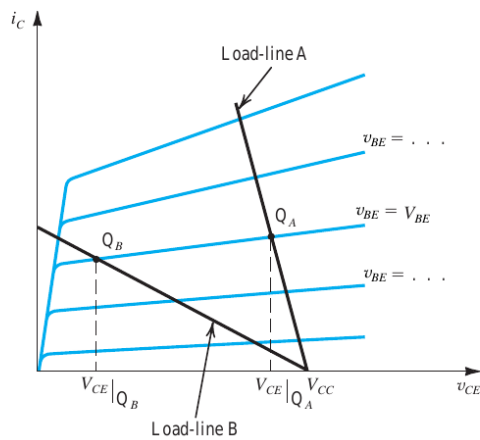
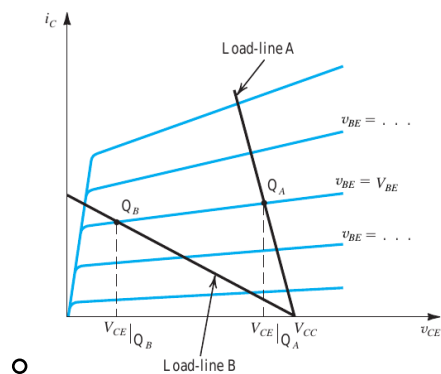


Figure 6.35 Effect of bias-point location on allowable signal swing: Load line A results in bias point Q_A with a corresponding V_{CE} that is too close to V_{CC} and thus limits the positive swing of v_{CE} . At the other extreme, load line B results in an operating point, Q_B , too close to the saturation region, thus limiting the negative swing of v_{CE} .

- Point Q_B is very close to the boundary of the saturation region,
 - thus severely limiting the allowable negative signal swing of v_{ce}
 - Exceeding this limit results in the transistor entering the saturation region,
 - for part of each cycle near the negative peaks,
 - resulting in a distorted output signal
 - we say the circuit doesnot have sufficient “legroom”



○